



GOSFORD HIGH SCHOOL

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2008

YEAR 12 HALF YEARLY HIGHER SCHOOL CERTIFICATE

MATHEMATICS EXTENSION 2

General Instructions:

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen.
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question.

Total marks: - 85

- Attempt Questions 1 -4

Question 1.

a) Write $\sqrt{3} + i$ in modulus argument form and hence evaluate $(\sqrt{3} + i)^6$. **3**

b) Write down the conjugate of $a+ib$ and hence show that if $z = x + iy$ **2**
then $\frac{z + \bar{z}}{z\bar{z}}$ is real.

c) i) If $\left| \frac{z-1}{z+1} \right| = 2$ where $z = x + iy$, show that the equation of the locus of z is **2**
$$\left(x + \frac{5}{3} \right)^2 + y^2 = \frac{16}{9}$$

ii) Represent this locus on an Argand diagram and shade the region for which **3**
the inequalities $\left| \frac{z-1}{z+1} \right| \leq 2$ and $0 \leq \arg z \leq \frac{3\pi}{4}$ are both satisfied.

d) What is the maximum value of $|z|$ for $|z - 1 - i| \leq 2$ **2**

e) i) Find the stationary points and the asymptote(s) of the function **4**
$$f(x) = \frac{(x+1)^4}{x^4 + 1}$$

ii) Sketch this function labelling all essential features. **2**

iii) Use the graph to find the set of values of k for which $(x+1)^4 = k(x^4 + 1)$ **2**
has two distinct real roots.

iv) Redraw your sketch of $f(x)$ from part (ii) and use it to do a neat sketch of **2**
$$y = \frac{1}{f(x)}$$

Question 2.

- a) By using the substitution $x = u^2$ find $\int_0^{\frac{1}{4}} \frac{dx}{\sqrt{x} \cdot \sqrt{1-x}}$ 3
- b) Using the substitution $x = 5 \sec \theta$ find $\int \frac{\sqrt{x^2 - 25}}{x} dx$ where $5 < x$. 3
- c) By using partial fractions find $\int \frac{9x-2}{2x^2-7x+3} dx$ 3
- d) i) Show that the function $y = \frac{x}{\sqrt{x^2+16}}$ is increasing for all values of x 2
- ii) The region R is bounded by $y = \frac{x}{\sqrt{x^2+16}}$, the x axis and the line $x = 4$. Show by using the substitution $y = \sin \theta$ and the result $\int \sec x dx = \ln(\sec x + \tan x)$, that the volume generated by rotating R about the y axis is $16\pi(\sqrt{2} - \ln(\sqrt{2}+1)) \text{ units}^3$ 4

Question 3

- a) For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, find : 5
- i) The length of the axes.
 - ii) Its eccentricity.
 - iii) The coordinates of the foci.
 - iv) The equations of the directrices.
 - v) Sketch the ellipse showing the above features.
- b) i) Show that the equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at any point θ is $\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$. 3
- ii) The normal at any point P on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the x -axis and y -axis at A and B respectively and $OACB$ is a rectangle, O the origin. Find the coordinates of Q in terms of θ . 2

- c) P is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with centre ' O '. A line drawn from O , parallel to the tangent to the ellipse at P , meets the ellipse at Q . Prove that the area of the triangle OPQ is independent of the position of P . 5
- d) The point $P(a \sec \theta, b \tan \theta)$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with focus S , is such that the tangent at P , the latus rectum through S , and one asymptote are concurrent. Prove that SP is parallel to the other asymptote. 4
(you may assume the equation of the tangent at P is $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$)

Question 4.

- a) If $2+i$ is a root of $x^3 - 2x^2 - 3x + 10$, find the other two roots. 2
- b) If $P(x) = x^4 + 2x^3 - 12x^2 + 14x - 5$ has a triple root, find the roots of $P(x)$. 3
- c) If α, β, γ are the roots of the equation $2x^3 - 4x^2 - 3x - 1 = 0$, find
- | | | |
|--------------------------------------|-------------------------------------|---|
| i) $\sum \alpha$ | ii) $\sum \alpha\beta$ | 6 |
| iii) $\alpha^2 + \beta^2 + \gamma^2$ | iv) $\alpha^3 + \beta^3 + \gamma^3$ | |
- d) If α, β, γ are the roots of the equation $x^3 - x - 1 = 0$, find the cubic equation with roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ 2
- e) Find all values of z such that $z^4 + 1 = 0$. 2
- f) Show that the zeros of $P(x) = x^4 + x^2 + 1$ are included in the zeros of $x^6 - 1$. Hence factorise $P(x)$ over the real numbers. 3
- g) Solve $2z^4 + 3z^3 + 5z^2 + 3z + 2 = 0$. 3
- h) Find $\cos 4\theta$ in terms of:
- | | |
|--|---|
| i) $\sin \theta$ and $\cos \theta$. | 3 |
| ii) $\cos \theta$ alone. | 2 |
| iii) Hence solve $8 \cos^4 x - 8 \cos^2 x + 1 = 0$ for $0 \leq x \leq \pi$. | 3 |

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

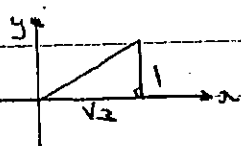
NOTE:

$$\ln x = \log_e x, x > 0$$

Ext 2 half yearly 2008 Solutions

Q1)

a) $\sqrt{3+i}$



$$|\sqrt{3+i}| = \sqrt{3+1} = 2$$

$$\arg(\sqrt{3+i}) = \tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6}$$

$$\therefore \sqrt{3+i} = 2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$$

$$\begin{aligned} (\sqrt{3+i})^6 &= \left(2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)\right)^6 \\ &= 2^6 (\cos \pi + i \sin \pi) \\ &= -2^6 \\ &= -64 \end{aligned}$$

b) conjugate = $a - ib$

$$\frac{z + \bar{z}}{z\bar{z}} = \frac{z + \bar{z}}{(x+iy)(x-iy)} = \frac{2x}{x^2+y^2}$$

which is real

Q ii) $\left| \frac{z-1}{z+1} \right| = 2$

$$\frac{|z-1|}{|z+1|} = 2$$

$$\frac{\sqrt{(x-1)^2+y^2}}{\sqrt{(x+1)^2+y^2}} = 2$$

$$\begin{aligned} \sqrt{(x-1)^2+y^2} &= 2\sqrt{(x+1)^2+y^2} \\ x^2-2x+1+y^2 &= 4(x^2+2x+1+y^2) \end{aligned}$$

$$x^2-2x+1+y^2 = 4x^2+8x+4+4y^2$$

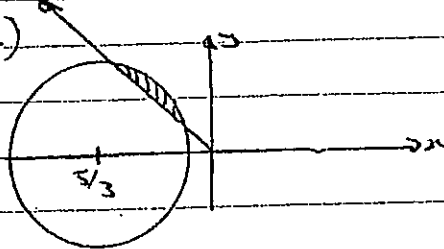
$$3x^2+10x+3y^2 = -3$$

$$x^2+\frac{10}{3}x+y^2 = -1$$

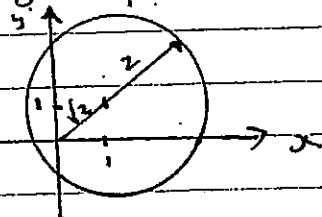
$$x^2+\frac{10}{3}x+\frac{25}{9}+y^2 = -1+\frac{25}{9}$$

$$(x+\frac{5}{3})^2+y^2 = \frac{16}{9}$$

ii)



Q1) $|z-1-i| \leq 2$



(circle centred at (1,1) rad = 2)
from diagram max
Value = $2 + \sqrt{2}$

e) $f(x) = \frac{(x+1)^4}{x^4+1}$

$$f'(x) = \frac{4(x^4+1)(x+1)^3 - 4x^3(x+1)^4}{(x^4+1)^2}$$

for s.p. $f'(x) = 0$

$$\Rightarrow 4(x^4+1)(x+1)^3 - 4x^3(x+1)^4 = 0$$

$$4(x+1)^3(x^4+1 - x^3(x+1)) = 0$$

$$4(x+1)^3(x^4+1 - x^4 - x^3) = 0$$

$$4(x+1)^3(1-x^3) = 0$$

$$x = -1 \text{ or } 1$$

$$y = 0$$

$$f'(-2) < 0$$

$$f'(0) > 0$$

$$f'(2) < 0$$

$$\therefore (-1, 0) \text{ min}$$

$$(1, 8) \text{ max}$$

asymptote

$$\lim_{x \rightarrow \infty} \frac{(x+1)^4}{x^4+1}$$

$$= \lim_{x \rightarrow \infty} \frac{x^4+4x^3+6x^2+4x+1}{x^4+1}$$

$$= \lim_{x \rightarrow \infty} \frac{4x^3+6x^2+4x}{4x^3+4x} = 1$$

$$= 1$$

$$\therefore \text{horizontal asymptote at } y = 1$$

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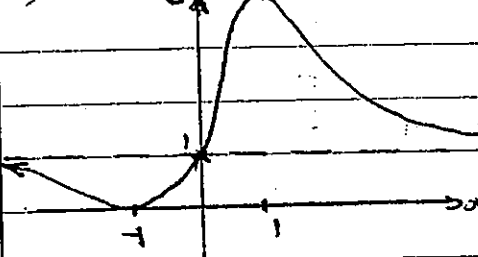
lim $x \rightarrow \infty$

$$\frac{x^4}{x^4} + \frac{4x^3}{x^4} + \frac{6x^2}{x^4} + \frac{4x}{x^4} + \frac{1}{x^4}$$

$$= 1$$

$\therefore$ horizontal asymptote at $y = 1$

ii)



iii) $(x+1)^4 = k(x^4+1)$

$$\frac{(x+1)^4}{x^4+1} = k$$

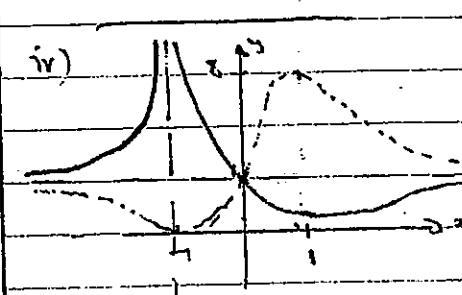
2 distinct real roots

$\Rightarrow$ horizontal line hits

the curve twice.

i.e. $0 < k < 1$ or $1 < k < 8$

iv)



Q2)

a) $\int_0^{\frac{1}{4}} \frac{dx}{\sqrt{x}\sqrt{1-x}}$

$$x = u^2 \quad x = 0 \Rightarrow u = 0$$

$$\frac{dx}{du} = 2u \quad x = \frac{1}{4} \Rightarrow u = \frac{1}{2}$$

$$dx = 2u du$$

$$\int_0^{\frac{1}{4}} \frac{2u du}{u\sqrt{1-u^2}}$$

$$= \int_0^{\frac{1}{2}} \frac{2 du}{\sqrt{1-u^2}}$$

$$= 2 \sin^{-1} u \Big|_0^{\frac{1}{2}} = 2 \sin^{-1} \frac{1}{2} = \frac{\pi}{3}$$

$$\frac{1}{2} \left[\sin^{-1} u \right]_0^{\frac{1}{2}}$$

$$\frac{1}{2} (\sin^{-1} \frac{1}{2} - \sin^{-1} 0)$$

$$\frac{1}{2} (\frac{\pi}{6} - 0)$$

$$= \frac{\pi}{12}$$

$$b) \int \frac{\sqrt{x^2-25}}{x} dx$$

$$x = 5 \sec \theta$$

$$\frac{dx}{d\theta} = \frac{5 \sin \theta}{\cos^2 \theta}$$

$$\int \frac{\sqrt{25 \sec^2 \theta - 25} \cdot \frac{5 \sin \theta}{\cos^2 \theta}}{5 \sec \theta} d\theta$$

$$= \int \frac{\sqrt{25 \tan^2 \theta} \cdot \frac{5 \sin \theta}{\cos^2 \theta}}{5 \sec \theta} d\theta$$

$$= \int \frac{\tan \theta \cdot \frac{5 \sin \theta}{\cos^2 \theta}}{\sec \theta} d\theta$$

$$= 5 \int \frac{\sin^2 \theta}{\cos^2 \theta} d\theta$$

$$= 5 \int \tan^2 \theta d\theta$$

$$= 5 \int (\sec^2 \theta - 1) d\theta$$

$$= 5 (\tan \theta - \theta)$$

$$\left(\begin{array}{l} x = 5 \sec \theta \\ \sec \theta = \frac{x}{5} \end{array} \right. \quad \begin{array}{c} \text{triangle} \\ \text{hypotenuse } x \\ \text{adjacent } 5 \\ \text{angle } \theta \end{array} \quad \begin{array}{c} \text{opposite } \sqrt{x^2-25} \end{array}$$

$$= 5 \frac{\sqrt{x^2-25}}{5} = \sec^{-1} \frac{x}{5} + c$$

$$= \sec^{-1} \frac{x}{5} + \sqrt{x^2-25} + c$$

$$c) \int \frac{9x-2}{2x^2-7x+3} dx$$

$$\int \frac{9x-2}{(2x-1)(x-3)} dx$$

$$\text{let } \frac{9x-2}{(2x-1)(x-3)} = \frac{a}{2x-1} + \frac{b}{x-3}$$

$$9x-2 = a(x-3) + b(2x-1)$$

$$\text{let } x=3: 25 = 5b$$

$$5 = b$$

$$\text{let } x = \frac{1}{2}: 2\frac{1}{2} = -2\frac{1}{2}a$$

$$-1 = a$$

$$\therefore \int \frac{9x-2}{2x^2-7x+3} dx = \int \frac{5}{x-3} - \frac{1}{2x-1} dx$$

$$= 5 \ln(x-3) - \frac{1}{2} \ln(2x-1)$$

$$= \ln \left(\frac{(x-3)^5}{\sqrt{2x-1}} \right) + c$$

$$d) i) f(x) = \frac{x}{\sqrt{x^2+16}}$$

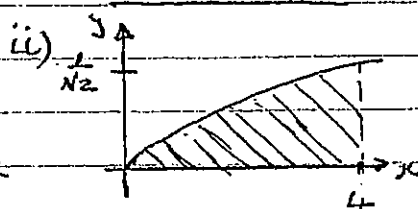
$$f'(x) = \frac{\sqrt{x^2+16} - \frac{x^2}{\sqrt{x^2+16}}}{x^2+16}$$

$$= \frac{x^2+16 - x^2}{\sqrt{x^2+16}^3}$$

$$= \frac{16}{\sqrt{x^2+16}^3}$$

which is positive for all values of x

$\therefore$ the curve is increasing for all values of x



$$V = \text{Volume of cylinder} = \pi \int_{-4}^4 x^2 dy$$

$$= \pi r^2 h = \pi \int_0^{\frac{1}{2}} \frac{16y^2}{1-y^2} dy$$

$$= \pi \times 4^2 \times \frac{1}{2} \quad y = \sin \theta$$

$$\frac{dy}{d\theta} = \cos \theta$$

$$dy = \cos \theta d\theta$$

$$= 8\pi(2 - \pi \int_0^{\frac{\pi}{4}} \frac{16 \sin^2 \theta \cdot \cos \theta d\theta}{1 - \sin^2 \theta})$$

$$= 8\pi(2 - \pi \int_0^{\frac{\pi}{4}} \frac{16(1 - \cos^2 \theta) \cos \theta d\theta}{\cos^2 \theta})$$

$$= 8\pi(2 - 16\pi \int_0^{\frac{\pi}{4}} \frac{1 - \cos^2 \theta}{\cos \theta} d\theta)$$

$$= 8\pi(2 - 16\pi \int_0^{\frac{\pi}{4}} \sec \theta - \cos \theta d\theta)$$

$$= 8\pi(2 - 16\pi [\ln(\sec \theta + \tan \theta) - \sin \theta]_0^{\frac{\pi}{4}})$$

$$= 8\pi(2 - 16\pi [\ln(\sqrt{2}+1) - \frac{1}{\sqrt{2}} - (\ln 1 - 0)])$$

$$= 8\pi(2 - 2\ln(\sqrt{2}+1) + \frac{1}{\sqrt{2}})$$

$$= 8\pi(2\sqrt{2} - 2\ln(\sqrt{2}+1))$$

$$= 16\pi(\sqrt{2} - \ln(\sqrt{2}+1))$$

$$Q3) \frac{x^2}{25} + \frac{y^2}{9} = 1$$

$$i) \text{ major axis} = 5$$

$$\text{minor axis} = 3$$

$$ii) b^2 = a^2(1 - e^2)$$

$$9 = 25(1 - e^2)$$

$$\frac{9}{25} = 1 - e^2$$

$$e^2 = \frac{16}{25}$$

$$e = \frac{4}{5}$$

$$iii) \text{ foci } (\pm ae, 0)$$

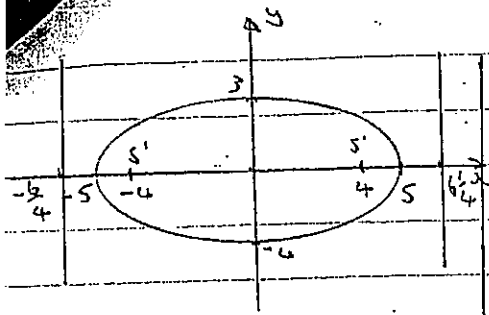
$$= (4, 0) \quad (-4, 0)$$

$$iv) x = \pm \frac{a}{e}$$

$$= \pm \frac{5}{\frac{4}{5}}$$

$$x = \pm \frac{25}{4}$$

$$x = \frac{25}{4}, \quad x = -\frac{25}{4}$$



$$b) i) \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{b^2 x}{a^2 y}$$

$$\therefore \text{Slope normal} = \frac{-a^2 y}{b^2 x}$$

$$\text{at } (a \sec \theta, b \tan \theta)$$

$$m = \frac{-a \tan \theta}{b \sec \theta}$$

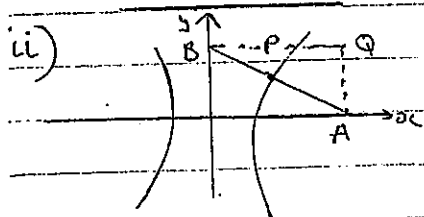
$$\therefore \text{eqn normal}$$

$$y - b \tan \theta = \frac{-a \tan \theta}{b \sec \theta} (x - a \sec \theta)$$

$$by \sec \theta - b^2 \tan \theta \sec \theta = -a \tan \theta + a^2 \sec \theta \tan \theta$$

$$a x \tan \theta + b y \sec \theta = \sec \theta \tan \theta (a^2 + b^2)$$

$$\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$$



ii)

$$\text{for A } (y=0)$$

$$\frac{ax}{\sec \theta} = a^2 + b^2$$

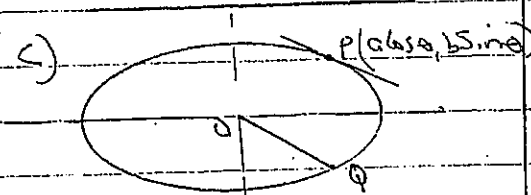
$$x = \frac{\sec \theta}{a} (a^2 + b^2)$$

$$\text{For B } (x=0)$$

$$\frac{by}{\tan \theta} = a^2 + b^2$$

$$y = \frac{\tan \theta}{b} (a^2 + b^2)$$

$$\therefore Q \left(\frac{\sec \theta}{a} (a^2 + b^2), \frac{\tan \theta}{b} (a^2 + b^2) \right) \text{ Now perpendicular distance P to OQ}$$



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \dots (1)$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-b^2 x}{a^2 y}$$

$$\text{at } (a \cos \theta, b \sin \theta)$$

$$\frac{dy}{dx} = \frac{-b \cos \theta}{a \sin \theta}$$

$$\therefore \text{eqn. OQ}$$

$$y = \frac{-b \cos \theta}{a \sin \theta} x$$

$$\text{Sub into (1)}$$

$$\frac{x^2}{a^2} + \frac{b^2 \cos^2 \theta x^2}{b^2 a^2 \sin^2 \theta} = 1$$

$$\frac{x^2}{a^2} + \frac{x^2 \cos^2 \theta}{a^2 \sin^2 \theta} = 1$$

$$\frac{x^2 \sin^2 \theta + x^2 \cos^2 \theta}{a^2 \sin^2 \theta} = 1$$

$$x^2 (\sin^2 \theta + \cos^2 \theta) = a^2 \sin^2 \theta$$

$$x^2 = a^2 \sin^2 \theta$$

$$x = a \sin \theta$$

$$\therefore y = \frac{-b \cos \theta}{a \sin \theta} \times a \sin \theta$$

$$= -b \cos \theta$$

$$\therefore Q (a \sin \theta, -b \cos \theta)$$

$$\therefore OQ = \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$$\text{eqn OQ: } b \cos \theta x + a \sin \theta y = 0$$

$$= \frac{b \cos \theta \cdot a \cos \theta + a \sin \theta \cdot b \sin \theta}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

$$= \frac{ab \cos^2 \theta + ab \sin^2 \theta}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$$

$$= \frac{ab (\cos^2 \theta + \sin^2 \theta)}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$$

$$= \frac{ab}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$$

$$\text{Now area of } \Delta OPQ$$

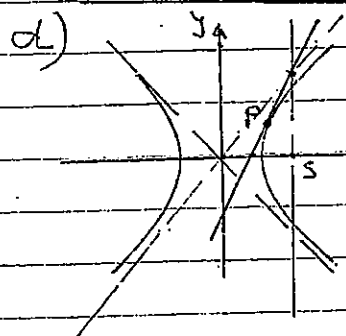
$$= \frac{1}{2} bh$$

$$= \frac{1}{2} \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$$\times \frac{ab}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$$

$$= \frac{ab}{2}$$

$$\therefore \text{Area independent of } \theta$$



$$\text{equation latus rectum}$$

$$x = ae \quad \dots (1)$$

$$\text{equation of asymptote}$$

$$y = \frac{b}{a} x \quad \dots (2)$$

$$\text{Sub (1) into (2)}$$

$$y = \frac{b}{a} ae$$

$$y = be$$

$$\therefore \text{pt. of intersection } (ae, be)$$

Now this point lies on the tangent

$$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$$

$$\frac{a \sec \theta}{a} - \frac{b \tan \theta}{b} = 1$$

$$e \sec \theta - e \tan \theta = 1$$

$$e(\sec \theta - \tan \theta) = 1 \quad \text{--- (A)}$$

Now gradient SP.

$$m = \frac{b \tan \theta}{a \sec \theta - e}$$

$$= \frac{b \tan \theta}{a(\sec \theta - e)}$$

from (A) $e = \frac{1}{\sec \theta - \tan \theta}$

$$m = \frac{b \tan \theta}{a(\sec \theta - \frac{1}{\sec \theta - \tan \theta})}$$

$$= \frac{b \tan \theta}{a(\frac{\sec^2 \theta - \sec \theta \tan \theta - 1}{\sec \theta - \tan \theta})}$$

$$= \frac{b \tan \theta}{a(\tan^2 \theta - \sec \theta \tan \theta)}$$

$$= \frac{b \tan \theta}{a(\tan \theta(\tan \theta - \sec \theta))}$$

$$= \frac{b \tan \theta}{a(-\tan \theta)}$$

$$= -\frac{b}{a}$$

$\therefore$ SP parallel to the other asymptote.

Q4)

a) $x^3 - 2x^2 - 3x + 10$

IF $2+i$ is a root then $2-i$ is a root (real coefficients)

Let the other root be α

$$\therefore \alpha + 2+i + 2-i = -\frac{b}{a} = 2$$

$$4 + \alpha = 2$$

$$\alpha = -2$$

$\therefore$ other roots; $2-i, -2$.

b) $P(x) = x^4 + 2x^3 - 12x^2 + 14x - 5$

$$P'(x) = 4x^3 + 6x^2 - 24x + 14$$

$$P''(x) = 12x^2 + 12x - 24$$

triple root $\Rightarrow$ root of $P(x), P'(x)$ and $P''(x)$

$$P''(x) = 0$$

$$12x^2 + 12x - 24 = 0$$

$$x^2 + x - 2 = 0$$

$$(x+2)(x-1) = 0$$

$$x = 1, -2$$

$$\text{Now } P'(1) = 4 + 6 - 24 + 14 = 0$$

$\therefore$ 1 root of $P'(x)$

$$P(1) = 1 + 2 - 12 + 14 - 5 = 0$$

$\therefore$ 1 root of $P(x)$

$$\therefore P(x) = (x-1)^3(x+5)$$

$\therefore$ roots are 1, 5

c) $2x^3 - 4x^2 - 3x - 1 = 0$

$$\therefore \sum \alpha = -\frac{b}{a} = 2$$

$$\text{ii) } \sum \alpha \beta = \frac{c}{a} = -\frac{3}{2}$$

$$\begin{aligned} \text{iii) } \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= 2^2 - 2 \times -\frac{3}{2} \\ &= 4 + 3 \\ &= 7 \end{aligned}$$

iv) $\alpha^3 + \beta^3 + \gamma^3$

if α a root then

$$2\alpha^3 - 4\alpha^2 - 3\alpha - 1 = 0 \quad \text{--- (1)}$$

if β a root then

$$2\beta^3 - 4\beta^2 - 3\beta - 1 = 0 \quad \text{--- (2)}$$

if γ a root then

$$2\gamma^3 - 4\gamma^2 - 3\gamma - 1 = 0 \quad \text{--- (3)}$$

$$(1) + (2) + (3)$$

$$2(\alpha^3 + \beta^3 + \gamma^3) - 4(\alpha^2 + \beta^2 + \gamma^2) - 3(\alpha + \beta + \gamma) - 3 = 0$$

$$2(\alpha^3 + \beta^3 + \gamma^3) - 4 \times 7 - 3 \times 2 - 3 = 0$$

$$2(\alpha^3 + \beta^3 + \gamma^3) - 28 - 6 - 3 = 0$$

$$2(\alpha^3 + \beta^3 + \gamma^3) = 37$$

$$\alpha^3 + \beta^3 + \gamma^3 = 18\frac{1}{2}$$

d) $x^3 - x - 1 = 0$

let root be x s.t. $x = \frac{1}{x} \therefore \alpha = \frac{1}{\alpha}$

Now α is a solution of $x^3 - x - 1 = 0$

$$\text{i.e. } \alpha^3 - \alpha - 1 = 0$$

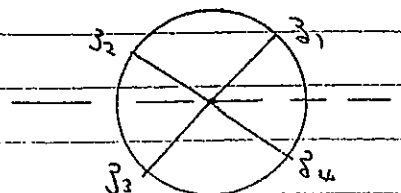
$$\left(\frac{1}{x}\right)^3 - \frac{1}{x} - 1 = 0$$

$$1 - x^2 - x^3 = 0$$

e) $z^4 + 1 = 0$

$$z^4 = -1$$

roots will be equally spaced around the unit circle starting at $\frac{\pi}{4}$



$$z_1 = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}(1+i)$$

$$z_2 = \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} = \frac{1}{\sqrt{2}}(-1+i)$$

$$z_3 = \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} = \frac{1}{\sqrt{2}}(-1-i)$$

$$z_4 = \cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} = \frac{1}{\sqrt{2}}(1-i)$$

$$f) P(x) = x^4 + x^2 + 1$$

$$x^6 - 1$$

$$= (x^2)^3 - 1$$

$$= (x^2 - 1)(x^4 + x^2 + 1) \quad \text{--- (A)}$$

$\therefore$ zeros of $P(x)$ are $(h)(\cos\theta + i\sin\theta)^4$

included in zeros of $x^6 - 1$

Now

$$x^6 - 1$$

$$= (x^3 - 1)(x^3 + 1)$$

$$= (x-1)(x^2+x+1)(x+1)(x^2-x+1)$$

$$= (x+1)(x-1)(x^2+x+1)(x^2-x+1)$$

equating this with (A)

$$\Rightarrow x^4 + x^2 + 1 = (x^2+x+1)(x^2-x+1)$$

$$z = \frac{-1 \pm \sqrt{1-16}}{4}, \frac{-1 \pm \sqrt{1-4}}{2}$$

$$= \frac{-1 \pm i\sqrt{15}}{4}, \frac{-1 \pm i\sqrt{3}}{2}$$

$$= \cos 4\theta + i\sin 4\theta$$

$$\text{also } (\cos\theta + i\sin\theta)^4$$

$$= \cos^4\theta + 4i\cos^3\theta\sin\theta - 6\cos^2\theta\sin^2\theta$$

$$- 4i\cos\theta\sin^3\theta + \sin^4\theta$$

$$= \cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta$$

$$+ i(4\cos^3\theta\sin\theta - 4\cos\theta\sin^3\theta)$$

equating real parts

$$\cos 4\theta = \cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta$$

$$\text{ii) } \cos 4\theta = \cos^4\theta - 6\cos^2\theta\sin^2\theta + \sin^4\theta$$

$$= \cos^4\theta - 6\cos^2\theta(1-\cos^2\theta) + (1-\cos^2\theta)^2$$

$$= \cos^4\theta - 6\cos^2\theta + 6\cos^4\theta + 1$$

$$- 2\cos^2\theta + \cos^4\theta$$

$$= 8\cos^4\theta - 8\cos^2\theta + 1$$

$$\text{iii) } 8\cos^4x - 8\cos^2x + 1 = 0$$

$$\Rightarrow \cos 4x = 0$$

$$4x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$$

$$x = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$$

$$\text{for } 0 \leq x \leq \pi$$

$$g) 2z^4 + 3z^3 + 5z^2 + 3z + 2 = 0$$

$$= z^2 \left(2z^2 + 3z + 5 + \frac{3}{z} + \frac{2}{z^2} \right) = 0$$

$$= z^2 \left(2 \left(z^2 + \frac{1}{z^2} \right) + 3 \left(z + \frac{1}{z} \right) + 5 \right) = 0$$

As $z=0$ is not a solution the solutions come from.

$$2 \left(z^2 + \frac{1}{z^2} \right) + 3 \left(z + \frac{1}{z} \right) + 5 = 0$$

$$2 \left(z + \frac{1}{z} \right)^2 - 4 + 3 \left(z + \frac{1}{z} \right) + 5 = 0$$

$$2 \left(z + \frac{1}{z} \right)^2 + 3 \left(z + \frac{1}{z} \right) + 1 = 0$$

$$\text{let } M = z + \frac{1}{z}$$

$$2M^2 + 3M + 1 = 0$$

$$(2M+1)(M+1) = 0$$

$$M = -\frac{1}{2} \text{ or } M = -1$$

$$z + \frac{1}{z} = -\frac{1}{2}, \quad z + \frac{1}{z} = -1$$

$$2z^2 + 2 = -z, \quad z^2 + 1 = -z$$

$$2z^2 + z + 2 = 0, \quad z^2 + z + 1 = 0$$